

DBMS (CS301)

Unit-wise Syllabus Breakdown

Unit 1 – Introduction to DBMS

Database concepts, file systems versus DBMS, database architecture, data models, schemas, and three-level architecture.

Unit 2 – Relational Model & SQL

Relational algebra, relational calculus, SQL queries, joins, views, constraints, and subqueries.

Unit 3 – Database Design

Entity-Relationship (ER) modeling, normalization, functional dependencies, and schema design.

Unit 4 – Transaction Management

Transactions, concurrency control, locking protocols, serializability, recovery techniques, and ACID properties.

Unit 5 – Advanced Database Concepts

Indexing, query optimization, NoSQL databases, distributed databases, data warehousing, and database security.